COMBINATORICS
ASSIGNMENT 2

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6. Prove that
$$P(n,k) = k!\binom{n}{k}.$$

Proof. First let's consider how we choose our k-permutations of $[n]$. We begin by choosing n elements, then $n-1$, then all the way down to $n-k+1$, so we see that
$$P(n,k) = n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$
On the right-hand side, we are choosing k-elements from $[n]$ in an unsorted fashion, but then we need to sort them in k ways which is $k!$, and we see that
$$k!\binom{n}{k} = k!\left(\frac{n!}{k!(n-k)!}\right) = \frac{n!}{(n-k)!}.$$
□

10. Prove that:
$$\sum_{k=0}^{n} r^{n-k}\binom{n}{k} = (r+1)^n$$

Proof. We'll go ahead and consider the general case here for $(x+y)^n$. We expand as:
$$\underbrace{(x+y)(x+y)\cdots(x+y)}_{n\text{-terms}}$$
which if we multiply out, we get 2^n terms each of which can be arranged as $x^{n-k}y^k$ for $k \in [n]$. We obtain the coefficient for the $x^{n-k}y^k$ term by choosing y in k of the n factors and x necessarily in the remaining $n-k$ factors. Thus the coefficient on $x^{n-k}y^k$ becomes the k-subsets of n factors or $\binom{n}{k}$. So we have
$$(x+y)^n = \sum_{k=0}^{n}\binom{n}{k}x^{n-k}y^k.$$
For the case in the problem we simply set $x = r$ and $y = 1$, and we are done. □

Date: February 9, 2026.

16. Prove Vandermonde's Identity:

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$$

Proof. The left-hand side counts all the k-element subsets of $[m+n]$. The right-hand side does the same thing according to the number of elements chosen from $[n]$. Indeed, we can first choose k elements from $[m]$ in $\binom{m}{k}$ ways, then choose the remaining $r-k$ elements from the set $\{m+1, m+2, \ldots, m+n\}$ in $\binom{n}{r-k}$ ways. □

19. Prove the identity:

$$\binom{n+1}{r+1} = \sum_{k=r}^{n} \binom{k}{r}$$

Proof. Let's go ahead and expand out the right hand side here so we see that:

$$\binom{n+1}{r+1} = \binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r}$$

Now with regards to the identity in question, the left-hand side clearly counts all the $r+1$-element subsets of $[n+1]$. And if we separate the right-hand side into cases according to the largest element. That is, there are $\binom{r}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+1$; there are $\binom{r+1}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+2$ and so on. In general there are $\binom{r+k}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+k+1$ for all $k \leq n-r$. □

20. Prove the identity:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k$$

Proof. We begin by noticing that the left hand side counts the non-empty subsets of $[n]$. To see the count of the right-hand side we count by classifying all the subsets of $A \subseteq [n]$ by each's largest element $m \in A$ where $m \in \{1, 2, \ldots, n\}$, $\max(A) = m$. We notice that every element of A is chosen from $[n]$, and elements $\{1, 2, \ldots, m-1\}$ can be included or not independently. So we have that:

$$2^n - 1 = \sum_{m=1}^{n} 2^{m-1}.$$

A simple shift in indices $k = m-1$ then results in:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k.$$

□

22. Prove the identity:

$$\binom{n}{k}\binom{k}{t} = \binom{n}{t}\binom{n-t}{k-t}$$

Proof. The left-hand side is the number of ways that we can choose S of k-elements from $[n]$, and then subsequetly choose t-elements from S. Regarding counting the right-hand side we see that we are immediately choosing t-elements from our original set of $[n]$. But then we are left with a set of $[n-t]$ from which we need to choose $k-t$ elements. So it works. □

24. Prove the identity:

$$\sum_{k=0}^{m}\binom{m}{k}\binom{n}{r+k} = \binom{m+n}{m+r}$$

Proof. Let A and B be disjoint sets with $|A| = m$ and $|B| = n$. We want to find the number of ways to chose $S \subseteq A \cup B$ with $|S| = m + r$, clearly $\binom{m+n}{m+r}$.

Now to find the right-hand side we count by spltting according to how many are chosen from A. For any such subset S, let $k = |S \cap A|$. We see the following

- $j \in \{0, \dots, m\}$, and
- once j is fixed we must choose:
  - j-elements from A: $\binom{m}{j}$ ways, and
  - the remaining $((m + r) - j)$-elements from B in $\binom{n}{(m+r)-j}$ ways.

So the total count follows as:

$$\sum_{j=0}^{m}\binom{m}{j}\binom{n}{(m+n)-j}$$

Now we re-index setting $k = m - j$ yielding $(m + n) - j = r + k$ with $\binom{m}{j} = \binom{m}{m-j} = \binom{m}{k}$, and sum becomes:

$$\sum_{k=0}^{m}\binom{m}{k}\binom{n}{m+k}$$

and we are done. □

26. Prove the identity:

$$\sum_{k=0}^{n} k^2\binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}$$

Proof. Begin with the binomial theorem for $(x + 1)^n$, which follows as:

$$(x+1)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$$

Now differentiate both sides, and we get

$$n(x+1)^{n-1} = \sum_{k=1}^{n} k \binom{n}{k} x^{k-1}$$

Now multiply both sides of the equation by x yielding:

$$nx(1+x)^{n-1} = \sum_{k=1}^{n} k \binom{n}{k} x^{k}$$

Differentiating again we get

$$n[(x+1)^{n-1} + (n-1)x(1+x)^{n-2}] = \sum_{k=1}^{n} k^2 \binom{n}{k} x^{k-1}$$

Now if we substitute in $x = 1$ we get

$$\begin{aligned} n[2^{n-1} + (n-1)x2^{n-2}] &= \sum_{k=1}^{n} k^2 \binom{n}{k} \\ n(n+1)2^{n-2} &= \sum_{k=1}^{n} k^2 \binom{n}{k} \end{aligned}$$

$\square$

33. Prove the identity:

$$\sum_{k=n-s}^{m-r} \binom{m-k}{r} \binom{n+k}{s} = \binom{m+n+1}{r+s+1}$$

Solution. $\square$